



Politecnico
di Torino

Algorithms and Programming

Iniziato martedì, 21 dicembre 2021, 17:13

Stato Completato

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Tempo impiegato 8 secondi

Valutazione 0,00 su un massimo di 14,00 (0%)

Domanda 1

Risposta non data

Punteggio max.:
2,00

The following recursive function is called by the main program with $n=3$, i.e., as $f(3)$.

```
int f (int n) {  
    if (n<=0) {  
        return (n);  
    }  
    return (f(n-1)+f(n-2)+f(n-3));  
}
```

Following the recursion tree, indicate the value returned by the function to the main program. Please, report your response as a unique integer value (with a sign, i.e., positive or negative). No other symbols must be included in the response. This is an example of the response format: -2

Risposta: ✖

La risposta corretta è : -7

Domanda 2

Risposta non data

Punteggio max.:
4,00

A list of lists represents the relationship between producers and products. Each producer's name is stored into an element of the main list and it has a related secondary list of products. Each product is characterized by an identifier (a string) and its price (a real value), and it is associated to an element of one secondary list.

Write the function to insert a new triple producer-product-price in this list of lists. Notice that the insertion procedure must discriminate among 3 cases:

1. The producer is not present in the database. In that case, the function must introduce the new producer with the new product and its price.
2. The producer is present but it does not have that product. In that case, the function must introduce the new product with its price in the secondary list associated with the right producer.
3. The producer and the product are already present. In this case, the function must update the price of that product for that producer.

The candidate can decide where to insert a new element (on top of the list, on the tail, or in a sorted position) for both the main and the secondary lists. The function must allocate all dynamically defined data fields.

The node data structure and the function prototypes are the following.

The function receives the reference to the pointer to the list of lists (head), and the triple producer name (name), product identifier (id), and price (price) to insert into the list of lists. Please, report the entire C code. Briefly explain (in plain English language, within a C comment) the logic of the code.

```
typedef struct producer_s producer_t;
typedef struct product_s product_t;
struct producer_s {
    char *name;
    producer_t *next;
    product_t *right;
};
struct product_s {
    char *id;
    float price;
    product_t *next;
};
```

```
void insert (producer_t **head, char *name, char *id, float price);
```

A large, empty rectangular box with a thin black border, intended for writing code or text. It occupies the upper half of the page.

Domanda 3

Risposta non data

Punteggio max.:
8,00

A palindrome string is one that reads the same backward as well as forward. For example, the strings "aba" and "123a321" are palindrome. A palindrome phrase is a set of strings and spaces that reads the same backward as well as forward once all the spaces have been eliminated and all capital letters have been transformed into small letters.

For example "was it a car or a cat I saw", "never odd or even", "amore roma", are palindrome phrases.

A file stores a set of strings, one string for each line. The number of strings is reported on the first line of the file. The length of each string is limited to 10 characters.

Write function **palindrome_phrase** which receives the name of the file and displays, on standard output, the longest palindrome phrase it can build using all strings stored in the file.

The prototype of the function is the following one:

```
void palindrome_phrase (char *name);
```

Notice that function **palindrome_phrase** can act as a wrapper and call other (possibly) recursive functions. Please, report the entire C code of the function. Briefly explain (in plain English language, within a C comment) the logic of the code and the combinatorics principle that has been adopted.
